SPAGHETTILAB | ONE BOARD, MANY POSSIBILITIES

Connect a circuit, choose how each port behaves, and explore what happens.

CONNECTIONS & POWER

CONTROL & LAB TOOLS

U14A–U14G are seven sections of ONE physical AG32 chip.

OPTIONAL ESP32 MODULE
ESP32-S3 | Optional wireless module
Adds Wi-Fi, Bluetooth and ESP32 project support.
DIRECT ACCESS
MAIN POWER
TPS562201 | Main power supply
Creates the power used by the board.
PORT LIGHTS
ACCESSORY POWER
ME6211 | 2.8 V accessory supply
Powers accessories that need a lower voltage.
ME6211 | 1.5 V accessory supply
Powers accessories that need a lower voltage.
OPTIONAL ACCESSORY LINKS
QWIIC ACCESSORIES
RIBBON–CABLE EXPANSION
BOARD–TO–BOARD LINK
G3VM | Network end switch
Enables the end-of-network connection when needed.
NETWORK CONNECTION
SN65HVD230 | Network interface
Connects this board to the shared CAN network.
USB CONNECTION
SIGNAL CONNECTIONS
U12 / U13 | Bidirectional signal switches
Connect or disconnect the ESP32 from the experiment ports.
SIX EXPERIMENT PORTS
Q3–Q8 | Electronic power switches
Give a port power or ground when selected.
PORT SUPERVISOR
AG32 section A | Experiment signals
Reads the ports and handles digital signals.
PORT MODE CONTROL
AG32 section B | Port settings
Chooses signal, power or ground for each port.
SUPERVISOR START–UP
20 MHz | Controller clock
Keeps the controller running at a steady pace.
AG32 section F | Power and start–up
Starts the controller and provides programming access.
COMMUNICATION HUB
AG32 section C | Board communication
Exchanges data with the computer and other boards.
CURRENT MEASUREMENT
AG32 section D | Current scan control
Chooses which port is measured.
4051 | First measurement selector
Connects one side of the selected current sensor path.
4051 | Second measurement selector
Connects the other side of the same sensor path.
INA226 | Current sensor
Measures the current of the selected port.
LOCAL POWER SUPPORT
ADJUSTABLE TEST OUTPUT
AG32 section E | Test signal control
Sets the test voltage and chooses its destination.
4051 | Test output selector
Sends the test signal to one of the six ports.
ESP32 PROGRAMS THE AG32
Loads the board controller without an external programmer.
GPIO47 TX / GPIO48 RX | GPIO15 BOOT0 | GPIO16 RESET (open–drain)
AG UART0: TX pin68 / RX pin69. BOOT1 stays LOW via R84.
Recovery: keep ESP32 pins Hi–Z; use UART pads or the existing JTAG pads.
SERVICE & SPARE ACCESS
AG32 section G | Spare inputs and port control
Spare analog inputs; ISO4/ISO5 use pins35/36.

Adds Wi-Fi, Bluetooth and support for ESP32 projects.

Connects accessories directly to the optional ESP32.

Turns the incoming power into the supply used by the board.

Shows the state of each port with coloured light.

Provides the supply voltages used by supported accessories.

Adapts selected connections for a supported accessory.

Connects compatible sensors with a small cable.

Connects displays, cameras and other supported add–ons.

Passes power and communication to neighbouring boards.

Lets this board exchange messages with the other boards.

Connects the board to a computer.

Connects or separates the ESP32 signals from the experiment.

Gives each port a signal, power or ground connection.

Observes the ports and manages digital signals for the experiment.

Chooses how each experiment port is connected.

Powers, starts and provides access to the board controller.

Links the controller to USB, the network and the optional ESP32.

Selects one port at a time and measures the current flowing through it.

Keeps the controller supply steady while it works.

Sends a chosen voltage to one port for experiments and simple component tests.

Provides spare analog inputs and two port–control test points.

Existing circuit preserved. Tester features require firmware and prototype validation; no new tester components were added.

Single–page schematic. Functional descriptions are not performance specifications.

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